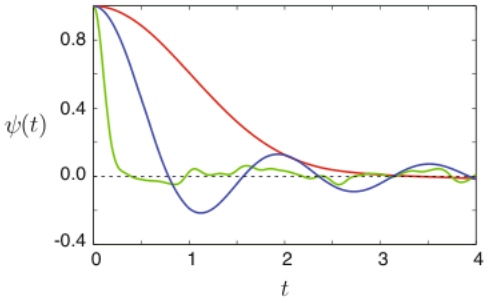



**Fig. 6.4** The memory kernel $\psi$ is shown for three different sets of frequencies $\Omega_i$. *Blue*: $k = 400$, $\Omega_i = 0.01i$; *red*: $k = 1{,}000$, $\Omega_i \sim \mathcal{U}[0,1]$; *green*: $k = 1{,}000$, $\Omega_i = 10\hat{\Omega}_i$, where $\hat{\Omega}_i \sim \mathcal{U}[0,1]$

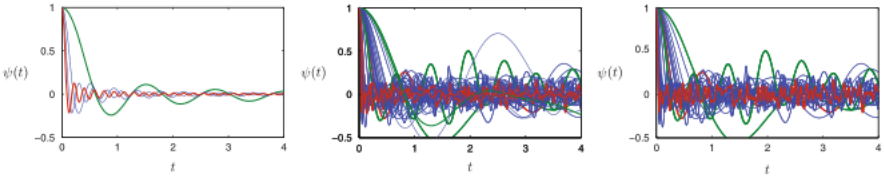


**Fig. 6.5** The memory kernels are shown for various choices of $\Omega_i^{(k)}$. *Left*: $\Omega_i^{(k)} = i/\sqrt{k}$, $k = 5^2, 10^2, \ldots, 35^2$; *center*: $\Omega_i = \sqrt{k}\hat{\Omega}_i$, $\hat{\Omega}_i \sim \mathcal{U}[0,1]$, $k = 10, 20, \ldots, 100$; *right*: $\Omega_i = k\hat{\Omega}_i$, $\hat{\Omega}_i \sim \mathcal{U}[0,1]$, $k = 10, 20, \ldots, 100$. In each case the *red curve* indicates the largest value of $k$ considered, whereas *green* indicates the smallest value

for example choosing them to be equally spaced on $[0, k]$, uniformly (randomly) distributed on a fixed interval, or uniformly distributed in an interval of length $\sqrt{k}$. Different choices of sequences $\Omega_i^{(k)}$ will lead to different appearances of the kernel $\psi$, as shown in Fig. 6.5.

If sufficient high frequency components are present, then the mass of the memory kernel is typically found to be localized around 0; in such a situation of rapid decay it is found that one may replace the function $\psi$ by a Dirac delta function scaled by a positive coefficient $\gamma$; we say that the system is Markovian or memory-less in this situation and we replace the delay-differential equation (6.29) by the SDEs

$$\mathrm{d}Q = P\mathrm{d}t, \tag{6.30}$$

$$\mathrm{d}P = -U'(Q)\mathrm{d}t + \sigma\mathrm{dW} - \gamma P\mathrm{d}t, \tag{6.31}$$

where we have viewed the function $f(t)$ as being approximated by a Wiener increment. This system is referred to as Langevin dynamics. The details of the rigorous derivation of (6.30)–(6.31) are beyond the scope of this book; consult the following [13, 113, 138, 150, 185, 198, 205, 350, 402]. We will consider computations of *generalized Langevin dynamics* in Chap. 8. For the moment we